

# Lesson 5 - Practical Application Guide

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## □ How to Run the Project

```
cd D:\learnlaravel2025\lessons\lesson-05\practice-app

# Setup database
copy .env.example .env
php artisan key:generate

# Configure database in .env
# DB_CONNECTION=mysql
# DB_DATABASE=lesson05_db

# Run Migrations
php artisan migrate

# Run Seeders
php artisan db:seed

# Start server
php artisan serve
```

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## □ Implemented Migrations

1. Categories Table
  2. Products Table (with Foreign Key)
  3. Orders Table
  4. Order Items Table
  5. Pivot Table: Product\_Tag
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## □ Implemented Seeders

### 1. CategorySeeder

### 2. ProductSeeder (with Factory)

### 3. UserSeeder

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## □ What We Learned

### 1. Creating Migrations

- □ Simple tables
- □ Foreign Keys
- □ Indexes
- □ Soft Deletes

### 2. Column Types

- □ Numbers: integer, decimal
- □ Strings: string, text
- □ Dates: timestamp, datetime
- □ Special: enum, boolean

### 3. Database Seeding

- □ Seeders for static data
- □ Factories for random data
- □ Organizing Seeders

### 4. Relationships

- □ One to Many (Category → Products)
- □ Many to Many (Products ↔ Tags)
- □ Foreign Keys with onDelete

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## □ Useful Commands

```
# Create migration
php artisan make:migration create_products_table

# Create Model with Migration
php artisan make:model Product -m

# Create everything
php artisan make:model Product -mfs
# -m: migration
# -f: factory
# -s: seeder

# Run migrations
php artisan migrate
php artisan migrate:fresh --seed

# Create seeder
php artisan make:seeder ProductSeeder

# Run specific seeder
php artisan db:seed --class=ProductSeeder
```

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## □ Testing Database

### Using Tinker

```
php artisan tinker
```

```
// Show all products
App\Models\Product::all();

// Show first 5 products
App\Models\Product::take(5)->get();

// Statistics
```

```
App\Models\Product::count();
App\Models\Category::count();
App\Models\User::count();

// Create record
App\Models\Category::create([
    'name' => 'New',
    'slug' => 'new',
]);
```

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## 📌 Tips

1. **Use SQLite for development** - Faster and easier
  2. **Write Seeders** for test data
  3. **Use Factories** for random data
  4. **Don't modify old migrations** - Create new one
  5. **Test migrations** before deployment
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## 📌 Next Step

After completing this lesson, you're ready for:

### **Lesson 6:** Eloquent ORM - Basics

- Creating Models
  - Querying data
  - Table relationships
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**Happy Learning!** 📖